Wycliffe Kipleting Rotich

Nairobi, Kenya | celestinerotich969@gmail.com
linkedin.com/in/rotichkipleting | github.com/wycliffRotich-dev/aethergrid
Open to remote roles | Available to start immediately

PROFESSIONAL SUMMARY

Platform Engineer with 7+ years building backend and distributed systems on cloud platforms, focused on concurrency, race-condition debugging, execution ownership, and failure recovery. Core stack: Python, FastAPI, PostgreSQL, AWS, Kubernetes, Kafka, Terraform, Docker, and React with TypeScript. Creator and maintainer of AetherGrid, an open-source distributed AI workload orchestrator covering scheduling, lease-based execution ownership, reconciliation, scoped API keys, hardened containers, and tenant-scoped data access. 696 passing tests and 68 documented architecture decisions.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL

Backend: FastAPI, REST APIs, Async Python, Pydantic, PostgreSQL, psycopg, SQLite, Node.js

Distributed Systems: Scheduling, Leases, Reconciliation, Concurrency, Race Condition Debugging, Retry Policies with Backoff and Jitter, Circuit Breakers, Idempotency, Worker Heartbeats, Node Draining, Process Management, Kafka

Cloud and DevOps: AWS, ECS, Kubernetes, Terraform, Docker, Docker Compose, Docker Secrets, GitHub Actions, CI/CD, Dependabot, Git

Security: API Key Authentication, Scoped Authorization, Tenant Isolation, Container Hardening (non-root, read-only filesystem, dropped capabilities)

Frontend: React 19, TypeScript, Vite, TailwindCSS, Client-Side Routing, Playwright

Testing: TDD, pytest, Contract Testing, Integration Testing, API Testing, End-to-End Testing, Mutation Verification

Design: Clean Architecture, Domain-Driven Design, SOLID, Aggregates, Entities, Value Objects, State Machines, Repository Pattern, Dependency Injection, Domain Events, Architecture Decision Records (ADRs)

AI and ML: LLM Evaluation, Code Quality Assessment, AI Training Pipelines, Data Annotation and QA

PROFESSIONAL EXPERIENCE

Platform Engineer (Independent) | AetherGrid, Open-Source AI Workload Orchestrator

Jan 2026 - Present

- Design and maintain AetherGrid from the domain model up, using Python, FastAPI, PostgreSQL (raw psycopg, no ORM), React, TypeScript, Docker Compose, and GitHub Actions. Four layers with a domain core that has no infrastructure imports. 696 passing tests (unit, contract, integration, API, end-to-end) and 68 ADRs.
- Built a multi-dimensional best-fit scheduler across CPU, memory, and VRAM, and time-bound leases renewed for the full run of a job. Reconciliation loops reclaim work from dead workers, expired leases, offline nodes, and jobs abandoned mid-cancellation, within a bounded retry budget.
- Found and closed a recurring race-condition class across lease release, outcome reporting, and lease renewal by fencing each on lease identity, then fenced reconciliation's own lease deletes by expiry and validity. Each fix was preceded by a test that reproduced the bug, and the fencing was verified against the real PostgreSQL repository.
- Built a real subprocess execution engine with enforced timeouts and two-stage termination (SIGTERM, then SIGKILL), and cancellation of running jobs delivered over the existing lease-renewal channel with an explicit CANCELLING status.
- Built a standalone worker agent that retries transient failures with capped exponential backoff and full jitter, re-registers after a control plane restart, and runs on one API key per agent so rate limits apply per agent. Verified by stopping and restarting the real API.
- Kept the API responsive during database outages: moved reconciliation off the event loop, split liveness from readiness, and bounded the shared connection pool timeout to 5 seconds. Worst-case latency on an unrelated endpoint dropped from about 29 seconds to 0.124 seconds, measured against a stopped Postgres container.

- Secured the API: key authentication on every route, scoped keys (jobs:execute to set a job command, keys:manage to issue and revoke keys), key ownership so a key can revoke only the keys it issued, and first keys minted only by a local script, never over HTTP. Verified by mutation: removing each check fails the test written for it.
- Hardened the containers: non-root user, read-only root filesystem, all Linux capabilities dropped, no-new-privileges, pinned image digests, and the database password as a Docker secret. A CI workflow asserts each property, and each check was shown to fail on a deliberate break. Dependabot covers docker, compose, actions, pip, and npm.
- Implemented tenant scoping for API keys, nodes, and jobs: the tenant comes from the credential, never from the request, repository methods either require a tenant or are named for across-tenants system use, and another tenant's resource answers like a missing one. Workers, leases, and events are the next stage.
- Built a resilient HTTP client with retries, a circuit breaker, OAuth2 and mTLS support, and replay of only requests that are safe to replay, verified over real HTTP that a POST applies its side effect exactly once.
- Built the React 19, TypeScript, Vite, and TailwindCSS operator dashboard with client-side routing, a live cluster event feed, and per-job event history. One shared data hook closed a stale-response race present in seven hooks, and a Playwright end-to-end test runs against the real API and Postgres in CI.

LLM Technical Trainer / AI Training Specialist | Mercor (Remote)

Aug 2025 - Present

- Evaluate LLM-generated code for correctness, architecture, concurrency, and API behavior.
- Review backend implementations where the happy path works but the system breaks under real conditions, identifying issues in state management, async execution, and race conditions.
- Write technical feedback that improves model reasoning and code quality.

AI Data Specialist, Video Annotation and Quality Review | Atlas Capture

Dec 2024 - May 2025

- Annotated and reviewed video datasets used in machine-learning workflows using CVAT, Labelbox, Scale, V7, and Encord.
- Performed QA reviews and resolved inconsistent or ambiguous annotations.

Software Engineer, Cloud Platforms | Toyota

Jan 2024 - Nov 2024

- Investigated and fixed concurrency problems in worker queues, resolving race conditions affecting Kafka-based ingestion pipelines for vehicle and telematics event data.
- Built high-volume data ingestion pipelines using Kafka for event streaming and asynchronous downstream processing.
- Deployed and orchestrated containerized workloads on Kubernetes and ECS across AWS environments supporting cloud-based vehicle and telematics services.
- Provisioned cloud infrastructure as code with Terraform, improving resource allocation and reducing provisioning time.
- Designed REST APIs with strict DTO validation before data reached downstream services, and refactored tightly coupled components to separate business logic from framework-specific code.

Backend Cloud Engineer | Kenya Revenue Authority (KRA)

Nov 2018 - Oct 2023

- Partnered with senior engineers on production revenue systems handling real financial data at national scale.
- Traced race conditions in code paths where concurrent requests could interfere with shared state, isolating root causes across application logic, shared state, and database interactions.
- Investigated timing-dependent bugs that were difficult to reproduce through standard request-response testing.

EDUCATION

Master of Science, Computer Science | United States International University-Africa (USIU-Africa), part-time | Jan 2022 - Dec 2024

Bachelor of Engineering, Computer Software Engineering | The Technical University of Kenya | Aug 2014 - Nov 2018